Akoizo Original Practice - Form 1 - Answer key

Division C | Hovercraft | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) TT is 10.0 s and the run takes 8.0 s. What is the Time Score?

A. 32

B. 8

C. 40

D. 20

Correct: A

2. (2 points) Five runs at one setting give 11.8, 12.4, 11.6, 12.5, and 11.9 s. What should the team improve first?

A. The Target Time

B. The dowel height

C. Repeatability, because the spread is almost 1 s

D. The calibration curve slope

Correct: C

3. (2 points) Four 1.2 V cells in series and two such strings in parallel give what voltage?

A. 2.4 V

B. 1.2 V

C. 4.8 V

D. 9.6 V

Correct: C

4. (2 points) Which contact with the track is NOT allowed?

Unit: Levitation: the air cushion and the skirt.
Worked-case data: A hovercraft with nickels weighs 1.6 kg and has a 0.35 m x 0.35 m cushion. What cushion pressure is needed?
Separate investigation observations:
Empty test: Vehicle glides freely across the floor with lift on.
With all nickels: Rear skirt drags and the craft stops quickly.
Nickels moved forward 6 cm: Glides freely again, sitting level.

A. The inside edge of the rails

B. Gentle skirt contact with the floor

C. The top of the side rails

D. The track surface

Correct: C

5. (2 points) When does the team's competition time end?

Unit: Impound, inspection, and competition day.
Worked-case data: A team's vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?
Separate investigation observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

A. After two complete or two incomplete runs, or 8 minutes, whichever comes first

B. After one complete run

C. After 10 minutes

D. When the team decides

Correct: A

6. (4 points) Calculate MS for a complete run carrying 2 rolls and 8 loose nickels.

answer: MS = 4 x 2 + 0.1 x 8 = 8.8. (4 points).

7. (4 points) Why does rail scraping hurt Time Score more than it hurts Distance Score?

answer: Scraping adds unpredictable friction that changes run time between runs, while the vehicle usually still finishes, so DS stays 40 but accuracy drops. (4 points).

8. (4 points) Explain why an early finish is not better than a late one.

Unit: Scoring a run: Distance, Time, and Mass.
Worked-case data: TT is 12.0 s. A vehicle carries 3 rolls and 10 loose nickels and finishes in 13.2 s with nothing falling off. Find the run score.
Separate investigation observations:
Run plan A: No nickels, typical error 0.3 s at TT = 12 s, always completes.
Run plan B: All nickels, typical error 0.6 s, completes 9 times in 10.
Run plan C: All nickels with ramp, error 0.2 s.

answer: TS depends on the absolute difference |t - TT|, so 1 s early and 1 s late give the same score. (4 points).

9. (4 points) How should a team use its first competition run to improve the second?

answer: Compare the first run time with TT and shift the setting along the calibration curve by the matching amount. (4 points).

10. (4 points) Why must a resistor used for speed control be rated for enough power?

answer: It turns the dropped voltage times the current into heat; an underrated resistor can overheat, change value, or burn out. (4 points).

11. (6 points) Solve this case and explain your reasoning: A hovercraft with nickels weighs 1.6 kg and has a 0.35 m x 0.35 m cushion. What cushion pressure is needed?

reasoning: Weight = 1.6 x 9.8 about 15.7 N.
Area = 0.35 x 0.35 = 0.1225 m^2.
P = 15.7 ÷ 0.1225 about 128 Pa. (3 points).

conclusion: About 128 Pa above atmospheric, which a small fan can supply easily if leaks are controlled. (3 points).

12. (6 points) Solve this case and explain your reasoning: A team’s vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?

reasoning: The dowel did not cross the start line within 3 seconds, so it is not a run.
The team still has two runs available while time remains.
Check the lift and thrust, restart quickly, and keep watch on the 8-minute clock. (3 points).

conclusion: A failed launch does not use a run, but it uses time, so diagnose fast. (3 points).

13. (2 points) What was the main cost of the open tray?

Supplied synthetic observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

A. The construction penalty on the whole run

B. The 0.3 Mass Score points from three nickels

C. Nothing, because the run was complete

Correct: A

14. (2 points) Define “Thrust” in the context of Hovercraft.

answer: Forward force from pushing air backward. (2 points).

15. (2 points) Define “Distance Score (DS)” in the context of Hovercraft.

answer: 40 for a complete run, prorated by distance for an incomplete one. (2 points).

16. (2 points) Define “Interpolation” in the context of Hovercraft.

answer: Estimating a setting between two measured points. (2 points).

17. (2 points) Define “Potentiometer” in the context of Hovercraft.

answer: Adjustable resistor used as a speed-setting knob. (2 points).

18. (2 points) Define “Hover gap” in the context of Hovercraft.

answer: Small opening between skirt and surface where air escapes. (2 points).